Solutions Manual
Chapter 1: Foundation of Engineering Economy

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Section 1: Cash Flows

Manufacturing Case Study

Exercise 1.1: Identifying Cash Inflows and Outflows

- **Bank Loan:** \$100,000 → **Cash Inflow (+)**
- **Equipment & Installation:** \$120,000 → **Cash Outflow (-)**
- **Annual Savings:** \$35,000 → **Cash Inflow (+)**
- **Annual Maintenance:** \$4,000 → **Cash Outflow (-)**
- **System Overhaul:** \$12,000 → **Cash Outflow (-)**
- **Salvage Value:** \$25,000 → **Cash Inflow (+)**

Exercise 1.2: Point vs. Range Estimates
Point Estimate: The guaranteed equipment cost of \$105,000.
Range Estimate: The installation cost (\$10,000 to \$15,000).
Advantage: Point estimates simplify calculations and make building mathematical models straightforward. **Disadvantage:** They can create a false sense of certainty and ignore risk.

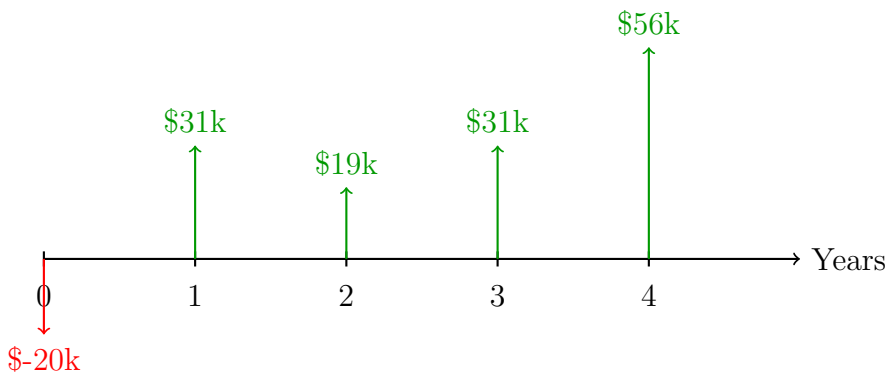
Advantage: Range estimates represent real-world uncertainty and risk more accurately. **Disadvantage:** They require more complex methods (like sensitivity analysis) to evaluate.

Exercise 1.3: Net Cash Flow Calculation (Initial Investment)
Year 0 Net Cash Flow = Inflows - Outflows
Year 0 = \$100,000 (Loan) - \$120,000 (Equip & Install) = **-\$20,000**

Exercise 1.4: Net Cash Flow Calculation (Operations)

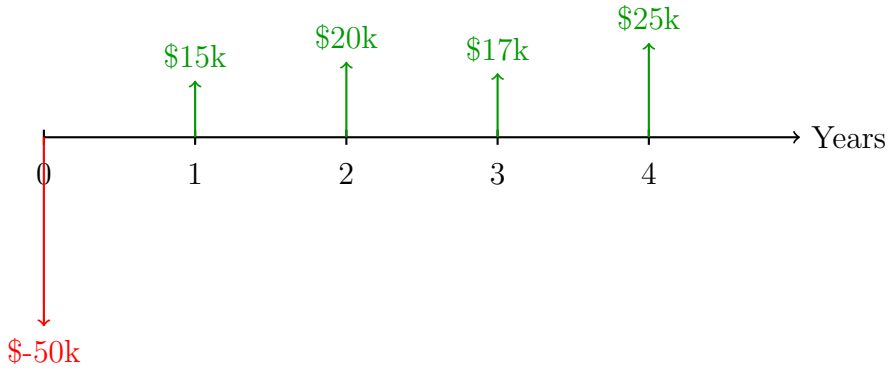
- **Year 1:** \$35,000 (Savings) - \$4,000 (Maintenance) = **+\$31,000**
- **Year 2:** \$35,000 - \$4,000 - \$12,000 (Overhaul) = **+\$19,000**
- **Year 3:** \$35,000 - \$4,000 = **+\$31,000**
- **Year 4:** \$35,000 - \$4,000 + \$25,000 (Salvage) = **+\$56,000**

Exercise 1.5: Drawing a Cash Flow Diagram



Independent Practical Scenarios

Exercise 1.6: Cash Flow Diagram - Software Upgrade
Net Cash Flows:
Year 0: **-$50,000**
Year 1: **+$15,000**
Year 2: **+$20,000**
Year 3: $25,000 - $8,000 = **+$17,000**
Year 4: $30,000 - $5,000 = **+$25,000**



Exercise 1.7: Cash Flow Profile of a Construction Project
Monthly Net Cash Flows:

- Month 0: **-$40,000**
- Month 1: **-$25,000**
- Month 2: **-$25,000**
- Month 3: -$25,000 + $60,000 = **+$35,000**
- Month 4: **-$10,000**
- Month 5: **-$10,000**
- Month 6: -$5,000 + $80,000 = **+$75,000**

(The cash flow diagram would feature downward arrows at months 0, 1, 2, 4, 5 and upward arrows at months 3 and 6).

Section 2: Simple and Compound Interest Calculations

Exercise 2.1: Simple vs. Compound Interest

Bank A (Simple): $F = P(1+ni) = 20,000(1+5\times0.085) = 20,000(1.425) = \$\mathbf{28,500}$

Bank B (Compound): $F = P(1+i)^n = 20,000(1+0.07)^5 = 20,000(1.40255) = \$\mathbf{28,051.03}$

Conclusion: Bank B offers the more economical deal, saving the engineer $448.97.

Exercise 2.2: Rate of Return and MARR

a) Interest = $54,500 - $50,000 = **$4,500**.
Rate of Return = $\frac{4,500}{50,000} \times 100 = \mathbf{9}\%$.

b) The MARR is 10%. Since the rate of return (9%) < MARR (10%), the investment fails to meet the required hurdle rate. It is mathematically not sound.

Exercise 2.3: Economic Equivalence

Find the Present Worth (PW) of the $54,000 payment next year at the MARR of 10%:
$P = \frac{F}{(1+i)^n} = \frac{54,000}{1.10} = \$\mathbf{49,090.91}$

Paying $54,000 next year is **equivalent** to paying $49,090.91 today when the MARR is 10%. Consequently, it is more economical to pay the supplier next year.

Note: What will really happen is that the company will invest the 50,000 at the MARR and gain $5,000. This interest amount can cover the difference ($4,000) and generate extra value ($1,000).

Exercise 2.4: Compound Interest over Multiple Periods

a) $F_4 = P(1+i)^n = 10,000(1+0.06)^4 = 10,000(1.262477) = \$\mathbf{12,624.77}$

b) Total Interest = $F_4 - P$ = $12,624.77 - $10,000 = $\mathbf{2,624.77}$

Exercise 2.5: The Rule of 72

a) Years to double $\approx \frac{72}{9} = \mathbf{8}$ years.

b) The project takes 8 years to double, but management requires it to double in 6 years. Therefore, it **fails** the initial screening.

Exercise 2.6: Deriving the Rule of 72

a) Set $F = 2P$ in the compound interest formula:
$2P = P(1+i)^n$
$2 = (1+i)^n$
Take the natural logarithm of both sides:
$\ln(2) = \ln((1+i)^n)$
$\ln(2) = n\ln(1+i)$
$n = \frac{\ln(2)}{\ln(1+i)}$

b) $\ln(2) \approx 0.693$. For small values of i, the Taylor series approximation gives $\ln(1+i) \approx i$. Therefore, $n \approx \frac{0.693}{i}$. If we express the interest rate as a percentage R (where $i = R/100$), then $n \approx \frac{69.3}{R}$.
Why 72 and why 5%-12%?
First, 72 is highly divisible (by 2, 3, 4, 6, 8, 9, 12), making mental math much easier than 69.3. Second, as interest rates increase, $\ln(1+i)$ starts to deviate from i. In the common interest rate range of 5% to 12%, using 72 in the numerator actually compensates for this deviation and yields a slightly more accurate result than 69.3. (For instance, at 8%, the exact doubling time is 9.006 years, and $72/8 = 9$ years, making it an extremely accurate approximation).

Exercise 2.7: The Rule for Simple Interest

a) Set $F = 2P$ in the simple interest formula:
$2P = P(1+ni)$
$2 = 1+ni$
$1 = ni$
$n = \frac{1}{i}$

b) If R is the percentage rate ($i = R/100$), then $n = \frac{100}{R}$. This is known as the **Rule of 100**.
For an 8% annual simple interest rate: $n = \frac{100}{8} = \mathbf{12.5}$ years.
Compared to compound interest (which doubles in 9 years), simple interest takes 3.5 years longer because the accumulated interest does not earn further interest.

Section 3: Introduction to Spreadsheet Functions

Exercise 3.1: Describing Financial Functions

1. Calculates the present value of a loan or an investment: `PV`
2. Calculates the future value of an investment: `FV`
3. Calculates the payment amount for a loan: `PMT`
4. Calculates the number of periods: `NPER`
5. Calculates the interest rate per period: `RATE`